

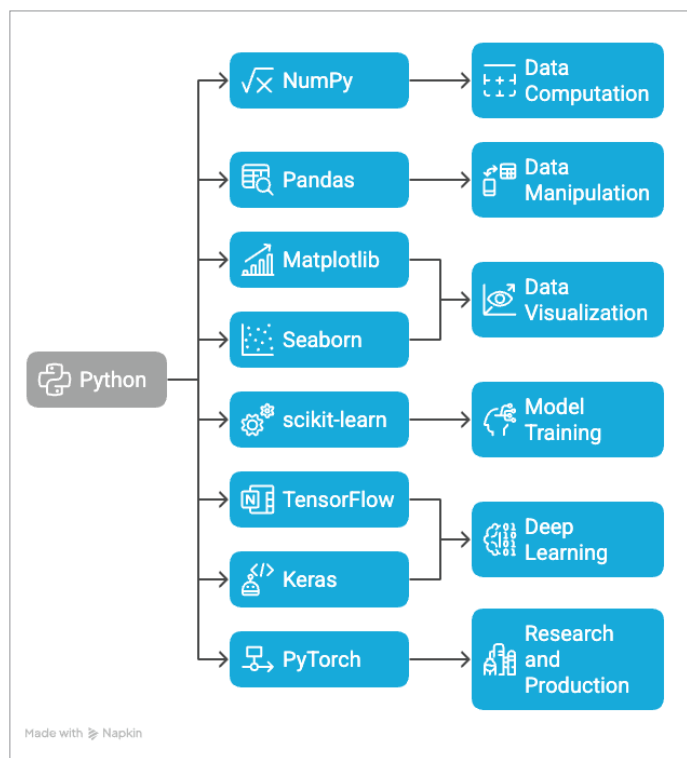


Figure 1: Python libraries for machine learning

Python libraries used in ML

A prime reason why Python is an unrivalled choice for machine learning is the powerful set of libraries associated with it. These libraries abstract complex mathematical operations into simple APIs for building, training, and deploying models.

NumPy: NumPy or Numerical Python is considered the fundamental package for scientific computation in Python. It supports high-performance multidimensional arrays and matrices, along with a few functions to operate on them. In machine learning, NumPy serves as the backbone for most computations ranging from vectorized operations to linear algebra.

Pandas: Pandas is the main library for data manipulation and analysis, and it introduces two core data structures, namely, Series and Data Frame, which make it simple to load and explore as well as clean and transform data. Etiquette handles missing values, filters out rows, aggregates statistics, etc, quite rapidly and intuitively. Pandas is a must in the first stages of any machine learning pipeline.

Matplotlib and Seaborn: Visualisation is paramount in trying to understand the patterns and relationships that reside within data. Matplotlib is a low-level plotting library, while Seaborn builds

Core Python concepts for machine learning

Concept	Description	Use in ML	Basic code example
Lists	Ordered, mutable collections of items	Store sequences of values like feature vectors or predictions.	<code>features = [0.2, 0.4, 0.9]</code>
Dictionaries	Key-value pairs for fast data lookup	Store dataset records, label mappings, or configuration parameters.	<code>labels = {'cat': 0, 'dog': 1}</code>
Tuples	Ordered, immutable collections	Represent fixed-size data like coordinates or hyperparameter sets.	<code>coords = (45.0, 90.0)</code>
Sets	Unordered collections of unique elements	Remove duplicates from data or compare feature sets.	<code>unique_words = set(word_list)</code>
Functions	Reusable blocks of code defined with <code>def</code>	Encapsulate repeated logic (e.g., data cleaning, metric calculation).	<code>def normalize(data): ...</code>
Loops	<code>for</code> and <code>while</code> loops to iterate through data	Iterate over datasets, apply transformations, or train over epochs.	<code>for x in data: process(x)</code>
List comprehensions	Compact syntax for generating lists from iterable objects	Efficiently transform or filter data in a single line.	<code>[x**2 for x in range(5)]</code>
Files	File handling with <code>open()</code> , <code>read()</code> , <code>write()</code>	Read datasets from text, CSV, or JSON formats.	<code>with open('data.csv') as f: ...</code>
Modules	Import reusable code from Python files or standard libraries	Organise code, use external libraries like NumPy, scikit-learn, etc.	<code>import pandas as pd</code>